Proof by Mathematical Induction

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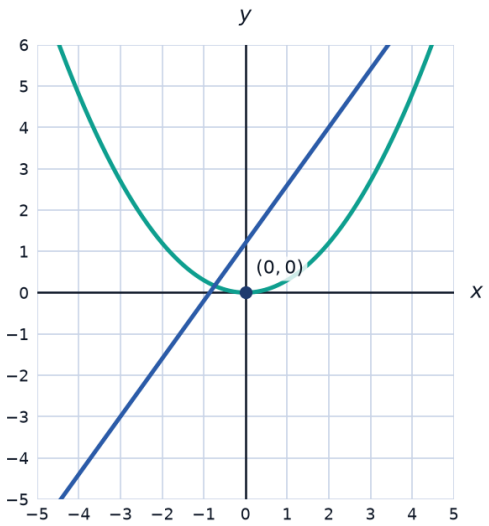
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The three steps of induction

State the three steps required for a valid proof by mathematical induction.

2 [Foundational] • Target: Proving a summation formula

Prove by induction that $1 + 2 + 3 + \cdots + n = n(n + 1)/2$ for all $n \geq 1$.



3 [Foundational] • Target: Proving divisibility results

Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

4 [Foundational] • Target: Induction with inequalities

Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.

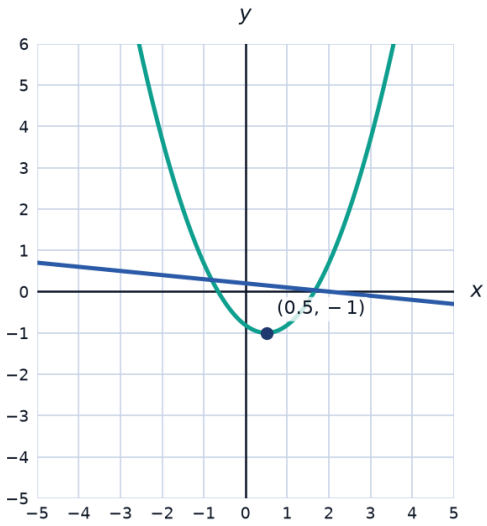
5 [Foundational] • Target: Induction in calculus: derivatives of a product

Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n.

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6 [Foundational] • Target: Induction with a recursively defined sequence

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.



7 [Foundational] • Target: Proving a geometric-series-style summation

Prove by induction that the sum $3^0 + 3^1 + \cdots + 3^{n-1}$ equals $(3^n - 1)/2$ for all $n \geq 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Proving a divisibility result with two variables

Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.

9 [Intermediate] • Target: Proving a divisibility result with two variables

Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.

10 [Intermediate] • Target: Induction with a recurrence relation for a sequence of sums

A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.

11 [Intermediate] • Target: A closing check: identifying an invalid induction argument

A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.

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12 [Intermediate] • Target: Proving a divisibility result by induction

Prove by induction that $n^3 - n$ is divisible by 6 for all integers $n \geq 1$.

13 [Intermediate] • Target: Induction in combinatorics: counting subsets

Prove by induction that a set with n elements has exactly 2^n subsets, for all integers $n \geq 1$.

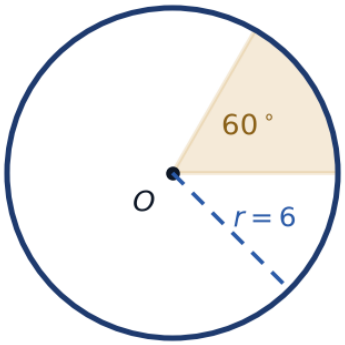
14 [Intermediate] • Target: A closing check: identifying an invalid induction argument

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Proving a sum-of-products formula by induction • Cross-Lesson: 6.01 Matrices (AI HL)

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.



16 [Challenging] • Target: Proving an exponential inequality by induction • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

17 [Challenging] • Target: Proving a divisibility result by induction • Cross-Lesson: 6.03 Markov Chains (AI HL)

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

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18 [Challenging] • Target: Proving a closed form for a recursive sequence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

[Algebra, Proof & Complex Numbers · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

19 [Challenging] • Target: A second exponential inequality by induction • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

20 [Challenging] • Target: A third inequality by induction • Cross-Lesson: 1.04 Partial Fractions

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n \to \infty} I_n$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The three steps of induction

In proving by induction that $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, assuming the result is true for $n = k$, what must be added to $\frac{k(k+1)}{2}$ to obtain the sum for $n = k + 1$?

a. k

b. $k + 2$

c. $k + 1$

d. $2k + 1$

22 [Foundational] • Target: Proving a summation formula

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. 100

b. 110

c. 10

d. 210

23 [Foundational] • Target: Proving divisibility results

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{4}}$ as a fraction.

a. $\frac{173}{160}$

b. $\frac{41}{32}$

c. $\frac{253}{160}$

d. $\frac{237}{160}$

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24 [Foundational] • Target: Induction with inequalities

For $x^3 - 19x - 30 = 0$, find the product of the three roots.

a. 30

b. 43

c. 27

d. 20

25 [Foundational] • Target: Induction in calculus: derivatives of a product

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45

c. 50

d. 54

26 [Foundational] • Target: Induction with a recursively defined sequence

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. 5

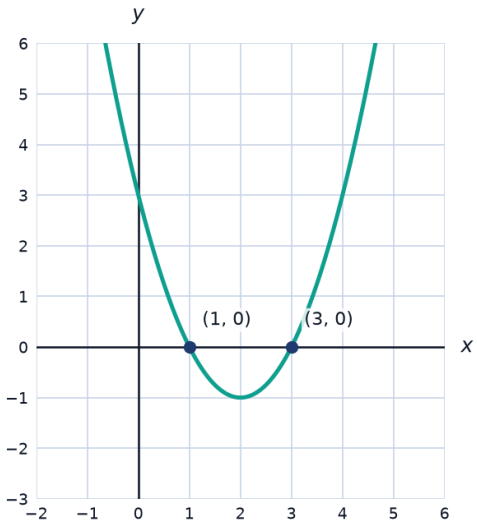
c. 6

d. 7

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27 [Foundational] • Target: Proving a geometric-series-style summation

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.



a. 4

b. 3

c. 1

d. -3

28 [Foundational] • Target: Proving an inequality involving factorials

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. 2

b. 0

c. -2

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Proving a divisibility result with two variables

Product of roots of $2x^2 - 5x + 3 = 0$:

a. $\frac{3}{2}$

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. 3

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30 [Intermediate] • Target: Induction with a recurrence relation for a sequence of sums

For $x^2 - x - 56 = 0$, find the product of the roots.

a. -42

b. -47

c. -56

d. -51

31 [Intermediate] • Target: Induction with a strengthened hypothesis

Simplify $(1 + 2i)(5 + 5i)$, giving your answer in the form $x + yi$.

a. $-3 + 16i$

b. $-5 + 15i$

c. $-1 + 12i$

d. $-3 + 14i$

32 [Intermediate] • Target: Induction to prove a formula for the sum of cubes

Write $z = 4(\cos 180^\circ + i\sin 180^\circ)$ in the form $x + yi$ (exact values).

a. $-4 + 1$

b. 4

c. -4

d. $-4i$

33 [Intermediate] • Target: Induction to prove a formula for a weighted geometric sum

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

a. 32

b. -1

c. $\frac{32}{3}$

d. 16

34 [Intermediate] • Target: A closing check: identifying an invalid induction argument

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$

b. $a_n = 3n + 2$

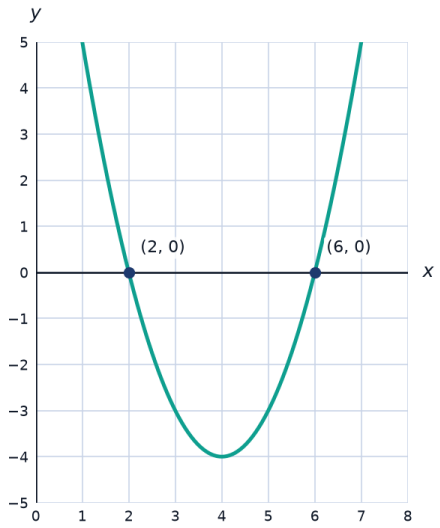
c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

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35 **[Intermediate]** • **Target: Proving a sum-of-products formula by induction**

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.



a. 12

b. 16

c. 20

d. 24

36 **[Intermediate]** • **Target: Proving an exponential inequality by induction**

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128

b. 96

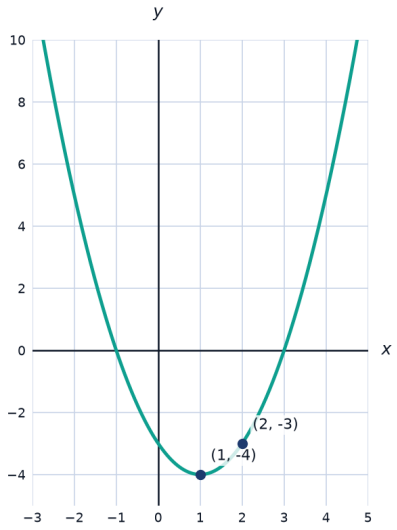
c. 74

d. 64

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37 **[Intermediate]** • **Target: Proving a divisibility result by induction**

The diagram shows the graph of the function $y = f(x)$. The function is of the form $f(x) = x^2 + bx + c$. The graph passes through the point $(2, -3)$ and has a minimum value of -4 at $x = 1$. Find the value of b.



a. -2

b. 1

c. 2

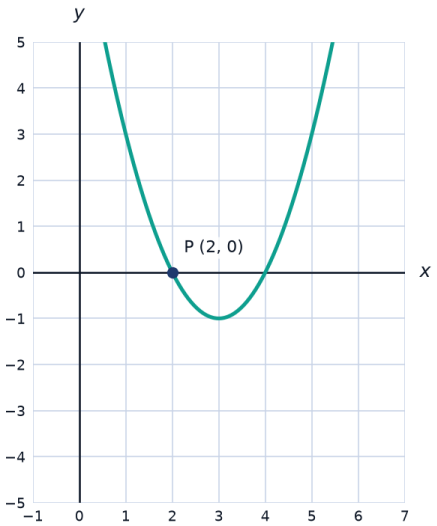
d. -1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging] • Target: Proving a closed form for a recursive sequence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.



a. 1

b. 2

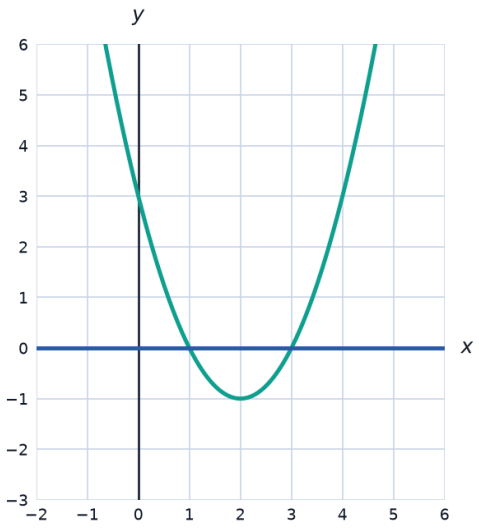
c. 4

d. 8

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39 **[Challenging]** • **Target: A second exponential inequality by induction** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

Consider the quadratic function $f(x) = x^2 - 4x + 3$. Find the value of x where the graph of $f(x)$ intersects the x-axis at the point $(x, 0)$, given that $x > 1$.



a. 2

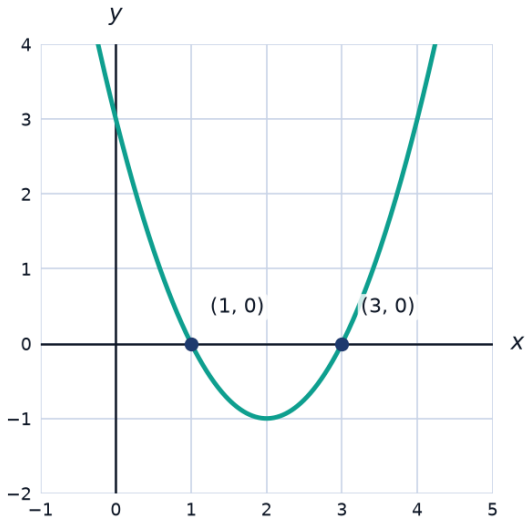
b. 3

c. 4

d. 5

40 **[Challenging]** • **Target: A third inequality by induction** • **Cross-Lesson: 1.04 Partial Fractions**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.



a. 1

b. 3

c. 4

d. -3

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41 [Challenging] • Target: Induction in combinatorics: counting subsets • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A geometric sequence has first term $a = 1$ and common ratio $r = 2$. Find the sum of the first 5 terms.

a. 31

b. -31

c. 32

d. 16

42 [Challenging] • Target: Proving the sum of the first n odd numbers • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Arithmetic series sum: S_{10} for $1 + 3 + 5 + \cdots$ (first 10 odd numbers):

a. 100

b. 55

c. 50

d. 99

43 [Challenging] • Target: The three steps of induction • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Consider the quadratic equation $x^2 - kx + 8 = 0$, where k is a real parameter. The graph of the corresponding quadratic function $y = x^2 - kx + 8$ is a parabola. Given that the parabola intersects the x-axis at exactly one point, find the value of k.

a. $-4\sqrt{2}$

b. $-2\sqrt{2}$

c. $2\sqrt{2}$

d. $4\sqrt{2}$

44 [Challenging] • Target: Proving a summation formula • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Let z_1 and z_2 be complex numbers such that $z_1 = 2 + i$ and $z_2 = 3 - 2i$. ; Consider the complex number $w = \frac{z_1^2}{z_2}$. ; If $w = a + bi$, where a and b are real numbers, find the value of $a^2 + b^2$.

a. $\frac{25}{13}$

b. $\frac{5}{13}$

c. $\frac{169}{25}$

d. $\frac{13}{5}$

45 [Challenging] • Target: Proving divisibility results • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Consider the complex number $w = 4 - 3i$. Find the modulus of w, denoted as $|w|$.

a. 5

b. 6

c. 7

d. 25

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